A STEP-BY-STEP GUIDE ON GETTING STARTED WITH ASSIGNMENTS ON VISUAL STUDIO

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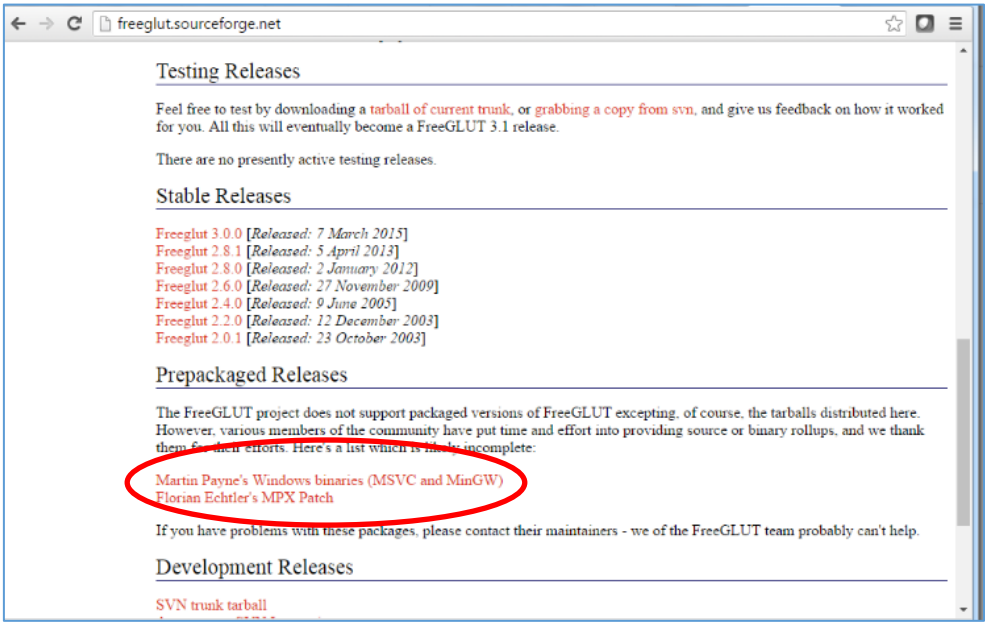
DOWNLOAD GLEW

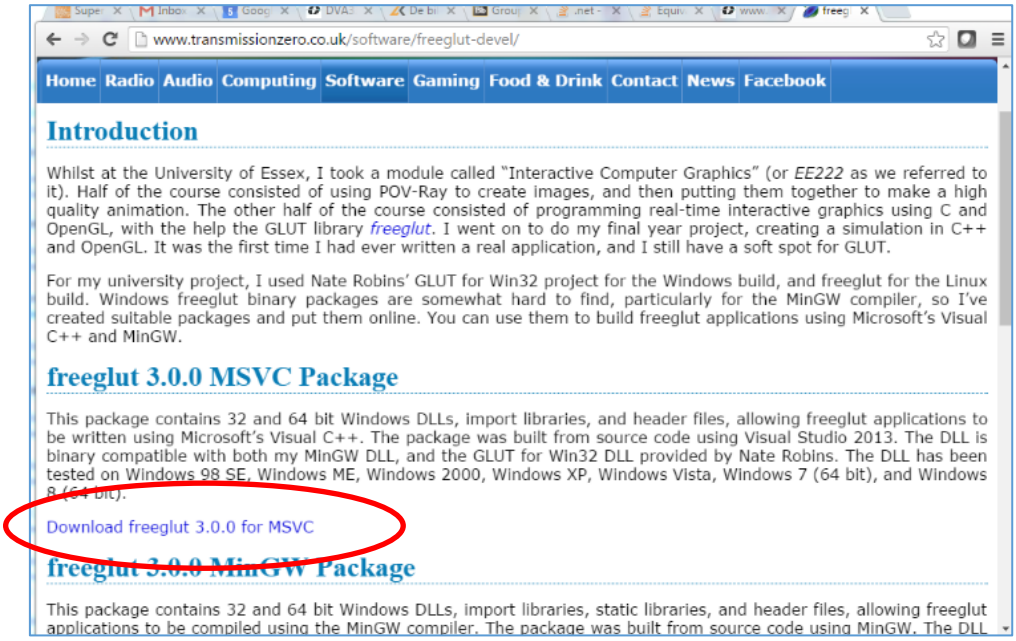
1. Go to http://glew.sourceforge.net/
2. Download **binaries** for ”Windows 32-bit and 64-bit”

DOWN LOAD FREEGLUT

1. Go to http://freeglut.sourceforge.net/
2. Scroll down until you see "Martin Payne's Windows binaries (MSVC and MinGW)". It is under the headline "Prepackaged Releases". Click the link.



3. Now download "freeglut 3.0.0 for MSVC"



UNPACK THE LIBRARIES

Pick a main development directory, say "DevOgl" and unpack the libs in a subdir called, for example, "Common". Check the directory structure. It will partly look something like this:

```
DevOgl
  Common
    freeglut
      bin
        x64
      include
        GL
      lib
        x64
    glew
      bin
        Release
          Win32
          x64
      doc
      include
        GL
      lib
        Release
          Win32
          x64
```

UNPACK THE STARTUP FILES

Unpack the startup files (DVA338_labsrc.zip) in "DevOgl" (<u>but NOT in common</u>). Rename to get this structure:

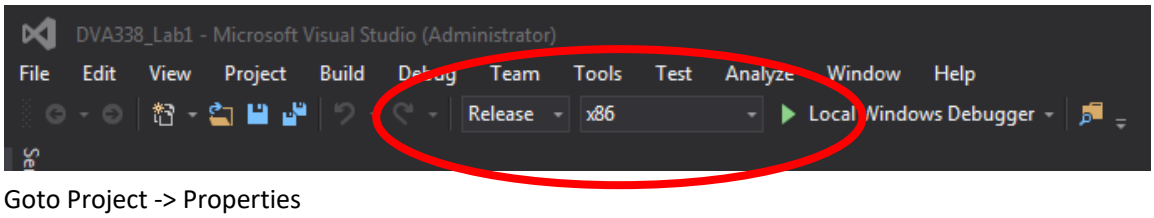
```
DevOgl
   Common
   DVA338_Lab1
      models
```

CREATE THE VS PROJECT

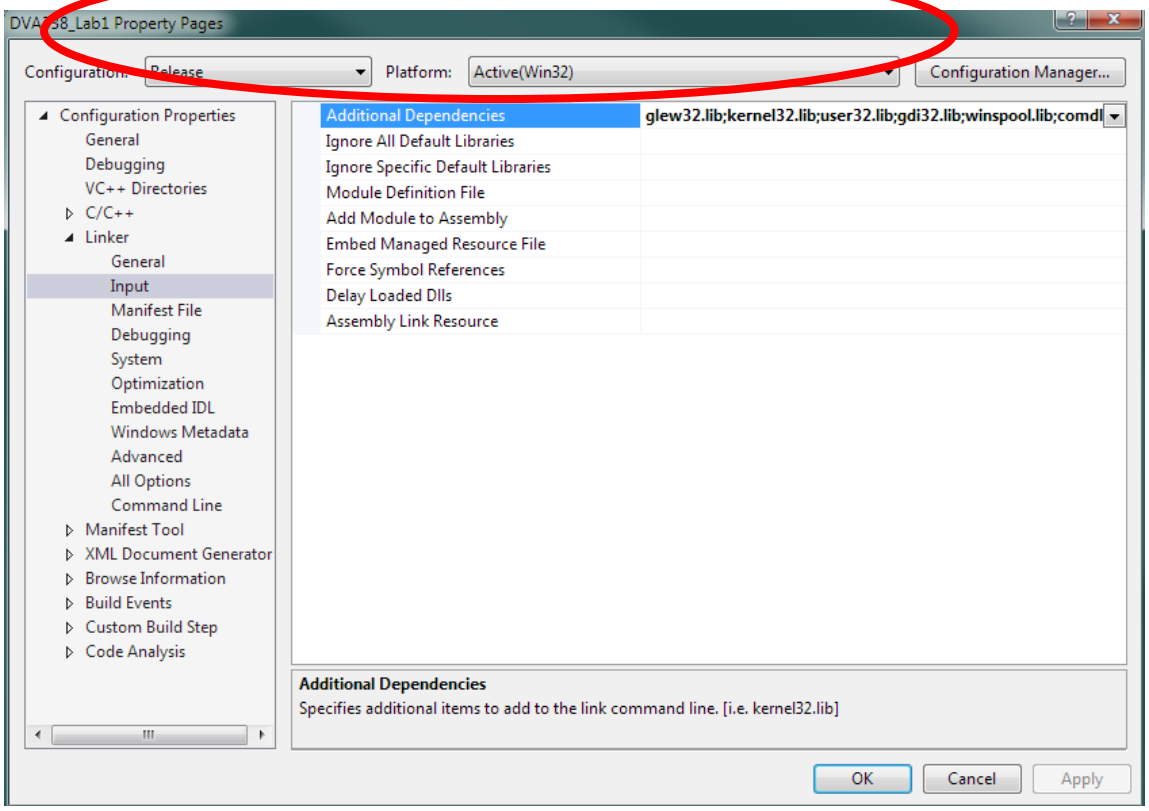
1. Start Visual Studio and create a new project of type "Empty Project" into the existing folder "DevOgl" (make sure that “Create directory for solution” is not checked). Choose the name of the project as the folder that you created for the lab (DVA338_Lab1). Press OK
2. Click “NEXT”
3. Select empty project. Uncheck the ”Precompiled Headers” and ”SDL” checkboxes (if they appear).
4. When the project is created, change the active configuration to "Release" mode.
5. Add the source files (the existing header and cpp files) to the project.
    a. In solution explorer right click on “Source Files” and choose “Add -> Existing item…”. Goto to “DVA338_Lab1” folder and select all the CPP files. Press ADD.
    b. In solution explorer right click on “Header Files” and choose “Add -> Existing item…”. Goto to “DVA338_Lab1” folder and select all the H files. Press ADD

SET DIRECTORY SETTINGS IN VS

1. On the toolbar make sure that “Release” and “x86” are selected.



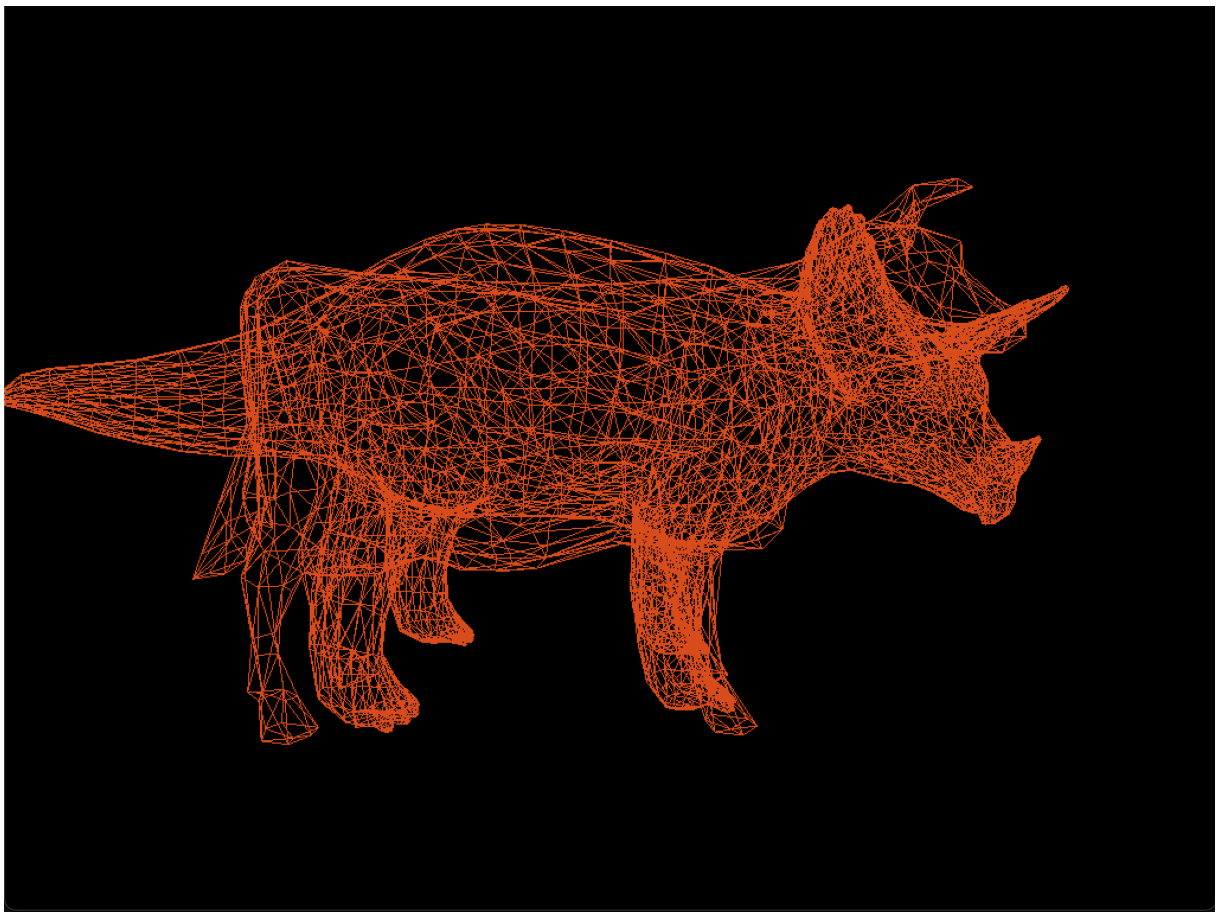
2. Goto Project -> Properties
3. On top of the dialog make sure that “Configuration” is set to “Release” and platform is set to “Win32”.



4. Specify the location of the header files "glew.h" and "freeglut.h". The setting is under "C++ -> General -> Additional Include Directories", once there type:
   `"..\Common\glew\include\GL;..\Common\freeglut\include\GL"`
5. Specify the location of the needed .lib files. The setting is under "Linker -> General -> Additional Library Directories", once there type:
   `"..\Common\glew\lib\Release\Win32;..\Common\freeglut\lib"`
6. **Note that your actual path to these directories might be different than the examples above.**
7. Also note that we are NOT using the x64 versions of the libraries.

FIX THE LIBRARY AND DLL DEPENDENCIES

1. Under "Linker -> Input -> Additional Dependencies", add "glew32.lib".
2. Copy the needed dll-files (freeglut.dll and glew32.dll) to your output directory, that is, "DVA338_Lab1\Release"
    a. If the “Release” directory does not exist, try running the code once. It will fail, but will create the folder for you ☺
    b. Again, note that you should not mix 32- and 64-bit libraries.

ENJOY

Compile and run the project. Hopefully, a window with the following rendering appears on your screen.